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EXAM AWS CERTIFIED AI PRACTITIONER AIF-C01 TOPIC 1 QUESTION 57 DIS...

Exam question from Amazon's AWS Certified AI Practitioner AIF-C01

Question #: 57

Topic #: 1

[All AWS Certified AI Practitioner AIF-C01 Questions]

A company has installed a security camera. The company uses an ML model to evaluate the security camera footage for potential thefts. The company has discovered that the model disproportionately flags people who are members of a specific ethnic group.

Which type of bias is affecting the model output?

A. Measurement bias

B. Sampling bias

C. Observer bias

D. Confirmation bias

Show Suggested Answer

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by Blair77 at *Nov. 12, 2024, 2:19 p.m.*

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Comments

65703c1 8 months, 1 week ago

Selected Answer: B

B is the correct answer

upvoted 1 times

Rcosmos 10 months, 3 weeks ago

Selected Answer: B

O viés que está afetando a saída do modelo é B. Viés de amostragem.

Esse tipo de viés ocorre quando os dados usados para treinar o modelo não representam adequadamente a população real. No caso da câmera de segurança, se o conjunto de dados de treinamento contiver mais exemplos de pessoas de um grupo étnico sendo associadas a atividades suspeitas, o modelo pode aprender padrões distorcidos e aplicar essas associações de forma desproporcional, levando a discriminação injusta.

Os outros tipos de viés não se aplicam diretamente a esse cenário:

upvoted 1 times

Willdoit 1 year, 1 month ago

Selected Answer: B

Sampling bias occurs when the data used to train the machine learning model is not representative of the entire population or the range of possible scenarios. If the model disproportionately flags people from a specific ethnic group, it suggests that the data used to train the model may have an overrepresentation or underrepresentation of certain ethnic groups, leading to biased predictions.

upvoted 4 times

Jessiii 1 year, 1 month ago

Selected Answer: B

B. Sampling bias occurs when the data used to train the model is not representative of the entire population. In this case, if the training data contains an overrepresentation or underrepresentation of certain ethnic groups, the model may disproportionately flag individuals from specific ethnic groups, leading to biased outcomes.

upvoted 2 times

pavankvv 1 year, 1 month ago

Selected Answer: A

Measurement bias occurs when the data used to train the machine learning model contains inherent biases or inaccuracies, leading to biased outputs or predictions.

In the given scenario, the security camera model is disproportionately flagging people from a specific ethnic group

upvoted 1 times

Moon 1 year, 3 months ago

Selected Answer: B

B: Sampling bias

Explanation:
Sampling bias occurs when the training data used for an ML model is not representative of the real-world population. In this case, the model disproportionately flags members of a specific ethnic group, likely because the training dataset was not balanced or representative of all

groups. This leads to skewed predictions that unfairly target certain populations.

upvoted 2 times

RightAnswers 1 year, 3 months ago

Selected Answer: B

Sampling bias: occurs when the data used to train the ML model is not representative of the overall population, leading to the model performing poorly on certain groups, like in this case where the model is disproportionately flagging people from a specific ethnic group.
Why the other options are not correct:
Measurement bias:
This refers to errors in the way data is collected or measured, which isn't directly related to the ethnic group bias in this scenario.
Observer bias:
This happens when a human observer's personal biases influence their interpretation of data, which isn't applicable here as the model is making the evaluations automatically.
Confirmation bias:
This refers to the tendency to seek out information that confirms existing beliefs, which isn't relevant to the training data used to develop the ML model.

upvoted 3 times

eesa 1 year, 3 months ago

Selected Answer: B

B. Sampling bias
Explanation:

Sampling bias occurs when the data used to train a model does not accurately represent the diversity of the population or real-world scenarios the model will encounter. In this case, if the training data for the security camera footage had an overrepresentation or underrepresentation of certain ethnic groups, the model may disproportionately flag members of that group as potential theft suspects. This leads to biased predictions due to imbalanced or unrepresentative training data.

upvoted 1 times

aws4myself 1 year, 4 months ago

Selected Answer: A

A. Measurement bias

Measurement bias occurs when the measurement process itself is flawed, leading to systematic errors. In this case, the model is likely biased due to the way it's trained on data that may not be representative of the entire population. This can lead to the model incorrectly associating certain characteristics with criminal behavior, particularly for individuals from underrepresented groups.

upvoted 1 times

Blair77 1 year, 4 months ago

Selected Answer: B

B - Sampling bias occurs when the data used to train a model is not representative of the population or real-world scenarios it's meant to

analyze. This leads to skewed results that favor or disfavor certain groups.

upvoted 4 times